

INFORMS 2026 Data Mining Society Data Challenge

Problem Description

This document describes the challenge problem, data, and submission requirements. It is intended for registered participants who have signed the Non-Disclosure Agreement (NDA) and been granted access to the Google Drive folder containing the data files. Read this document in full before opening the data.

Background

On March 13–19, 2026, two successive wind-driven storm systems swept through Indiana (IN), Ohio (OH), Pennsylvania (PA), and West Virginia (WV), causing widespread power outages across 302 counties. Hourly outage data at the county level was collected from PowerOutage.com for the full event window, together with a two-day pre-event lag period (March 11–12) included to support lag feature construction.

Power outages during storms impose significant burdens on utilities, emergency managers, and affected residents. Predicting how severe an outage will become over the coming hours and days, using only information available at the time of the prediction, allows utilities to pre-position repair crews, request mutual aid from neighbouring systems, and prioritise emergency response before conditions deteriorate. This challenge asks you to build models that make exactly these predictions from real storm data, learning cross-county outage dynamics from 239 fully observed training counties and applying them to 63 held-out test counties.

Data Overview

Event structure

- **Full window:** March 11–19, 2026 (9 days, 216 hours per county)
- **Pre-event window:** March 11–12 (baseline and lag signal)
- **First storm wave:** March 13–14
- **Second storm wave:** March 16–17
- **Coverage:** 302 counties across IN, OH, PA, WV
- **Weather:** Hybrid URMA (2.5 km, wind/temperature) and ERA5 (~31 km, atmospheric context)

Train / test structure

The split is county-level, not time-based. Train and test counties never overlap.

State	Train	Test	Total
IN	72	20	92
OH	70	18	88
PA	52	15	67
WV	45	10	55
Total	239	63	302

Training file (DM_Train.csv): 239 counties $\times$ 216 hours = 51,624 rows. All variables present for all hours. All OSI forecast targets included.

Test file (DM_Test.csv): 63 counties $\times$ 216 hours = 13,608 rows. Two types of columns:

- **Weather variables:** provided for all 216 hours. A real utility operator has access to weather forecasts for the full storm window; this mirrors that operational reality.
- **Outage variables** (outageCount, outage_pct, P_t , N_t , D_t , R_t , **osi**, all lag columns): provided only for March 11–13 (first 72 hours). **NaN for March 14–19.** OSI forecast targets withheld entirely.

This structure reflects a genuine operational forecasting scenario: you observe the first 72 hours of a storm event (pre-event baseline plus first wave onset) and must predict how outage severity will evolve over the remaining 144 hours, leveraging patterns learned from 239 fully observed training counties.

Outage Severity Index (OSI)

The OSI is a composite hourly severity score built from four variables:

Var	Type	What it captures
P_t	State	Current outage fraction (customers out / customers tracked)
N_t	Flow	New outage growth rate this hour (getting worse)
D_t	State	6-hour rolling mean of P_t (sustained persistence)
R_t	Flow	Restoration rate this hour (actively recovering)

$$\text{OSI}_t = 0.40 P_t + 0.35 N_t + 0.25 D_t - 0.10 R_t \quad (\text{clipped } \geq 0)$$

Full OSI documentation is in `osi_methodology.pdf`.

Prediction Task: Hourly OSI Forecasting

At each hour t in the March 14–19 prediction window, using outage information from the March 11–13 observed window and weather information for any timestamp, predict the OSI at four future horizons:

Target	Horizon	Operational meaning
osi_target_t01h	$t + 1$ hour	Immediate next-hour severity
osi_target_t06h	$t + 6$ hours	Crew pre-positioning window
osi_target_t24h	$t + 24$ hours	Next-day mutual aid planning
osi_target_t48h	$t + 48$ hours	Two-day emergency outlook

Predictions required: One row per test county per hour for the March 14–19 prediction window (9,072 rows total; 63 counties $\times$ 144 hours). Rows where the horizon target extends beyond March 19 carry no ground truth and should remain NaN in your submission. Submit via `sample_submission.csv` (provided in your folder).

Temporal Causality Rule

At prediction origin t , the following applies:

Allowed	Any outage feature value at timestamp $\leq t$
Allowed	Weather features at any timestamp (present and future)
Forbidden	Any outage feature value at a timestamp later than t
Forbidden	Computing OSI or any outage-derived quantity from future rows

Enforcement: Code submissions will be reviewed for compliance. Any submission found to have used future outage values will be **automatically rejected**, regardless of predictive accuracy.

Submission Requirements

Each submission must include three components organised in a single folder named `TeamName_Submission/` and submitted via the Google Form link provided to registered participants.

Component 1: Prediction file

Use `sample_submission.csv` as the starting point. Fill in predicted OSI values for all four horizon columns. Do not alter identifier columns or row order.

Component 2: Reproducible code

Submit as `code.zip`. Must reproduce submitted predictions end-to-end on the provided data files with no manual steps. Include library versions, explicit random seeds, and relative file paths.

Component 3: Written report

No more than **6 pages** (excluding references), submitted as `report.pdf`. Describe the methodology, key modelling decisions, and a critical discussion of results. The report is a central part of the evaluation.

How to submit

Upload your `TeamName_Submission/` folder to your own Google Drive, share with view access, and submit the link via the Google Form provided to registered participants. Deadline: **September 25, 2026 (AOE)**.

Files in Your Google Drive Folder

File	Description
<code>DM_Train.csv</code>	51,624 rows $\times$ 63 columns. 239 counties, full 216-hour window, all variables and targets.
<code>DM_Test.csv</code>	13,608 rows $\times$ 43 columns. 63 counties, full 216-hour window. Outage variables present for March 11–13 only; weather present for all 216 hours; OSI targets withheld.
<code>sample_submission.csv</code>	Submission template. 9,072 rows (March 14–19, 63 counties $\times$ 144 hours). Fill in OSI predictions for the four horizon columns.
<code>problem_description.pdf</code>	This document.
<code>osi_methodology.pdf</code>	OSI formula, component definitions, and data quality notes.
<code>weather_variables.pdf</code>	Weather feature sources, units, and aggregation methods.
<code>variable_descriptions.pdf</code>	Description of every column in train and test files.

Note on severity_tier: Present in the training file for stratified cross-validation only. Must not be used as a model input feature. Absent from the test file.

Note on OSI reconstruction: In the March 11–13 observed window, all four OSI components (P_t , N_t , D_t , R_t) are present in the test file. OSI and its lags can be reconstructed from these components using the formula above, following the temporal causality rule. Full instructions are in `osi_methodology.pdf`.